Multi-task Learning of Negation and Speculation for Targeted Sentiment Classification

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Contributions:

1. Quantify the effects of negation and speculation for Targeted Sentiment Analysis (**TSA**). We find that all models perform on average **24** and **25** percentage points worse on the negation and speculation challenge datasets.
2. Two English challenge datasets annotated for negation and speculation.
3. Find both transfer and multi-task learning can create more robust models to the effects of negation and speculation.

Challenge Dataset

All targets in the negation (***Neg***) and speculation (***Spec***) challenge datasets have been annotated so that they contain the respective lingustic attribute. This annotation was performed for the SemEval **Laptop** and **Restaurant** datasets for the **development** and **test** splits.

The example sentence below shows how negation and speculation has been added:

Type	Sentence	Sentiment towards sushi
Original	This is good, inexpensive sushi.	Positive 😀
Negated	This is **not** good, inexpensive sushi.	Negative ☹️
Speculative	**I'm not sure if** this is good, inexpensive sushi.	Neutral 😐

Modelling

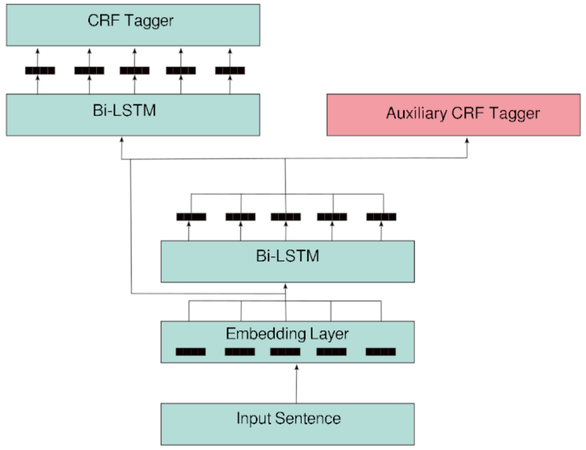
Embedding Layer Options:

1. GloVe - No Transfer Learning (**TL**)
2. Pre-Trained Transformer - **TL**

Embedding layer parameters do not get tuned (frozen parameters).

Single Task Learning (**STL**) only uses the Green layers.

Multi-task Learning use the Pink layer for the auxiliary task. Auxiliary and main task share all layers below the Pink layer.



Results - Challenge Dataset

F1 scores for the combined extraction and sentiment classification. Blue highlights are the results that are better than **STL**, **bold** indicates the best result.

			Auxiliary Task							
			NEG_{CD}	DR	LEX	NEG_{SFU}	SPEC	UPOS	STL	
Negation	Restaurant	GloVe	43.28	40.95	39.19	**45.22**	39.85	41.43	40.38	10.3
		TL	50.40	**51.92**	51.15	50.75	51.49	49.45	50.68	
	Laptop	GloVe	32.28	29.30	30.47	**33.96**	31.36	30.36	29.33	8.81
		TL	**39.95**	38.08	38.35	39.18	37.59	38.23	38.14	
Speculation	Restaurant	GloVe	47.14	**48.94**	48.07	47.90	48.41	47.94	48.35	4.59
		TL	52.39	53.69	**55.15**	53.25	52.34	52.55	52.94	
	Laptop	GloVe	25.67	26.39	**28.59**	27.33	26.95	27.31	26.01	2.71
		TL	28.36	28.09	28.56	28.33	28.54	28.37	**28.72**	

Future Directions

We find that **TL** and **MTL** do not always complement each other. Additionally certain auxiliary tasks improve different parts of the **TSA** subtasks e.g. **UPOS** improves extraction but not sentiment classification. Thus future work should consider a model that can better incorporate the auxiliary tasks with the **TSA** subtasks. This decoupling could also allow **MTL** and **TL** to complement each other more.

Code / data / models available

https://bit.ly/33yyRLI